

Solutions:

(1)

Qno. 04

As $n = pq$, and p and q are distinct, so 1, p , q , n are the only factors possible.

1, 8839, 8849, 78216311

Qno. 05

Scalability issues

Single point of failure in case of centralized solution, and in realm in case of distributed

Separation of concerns

Single sign on (SSO)

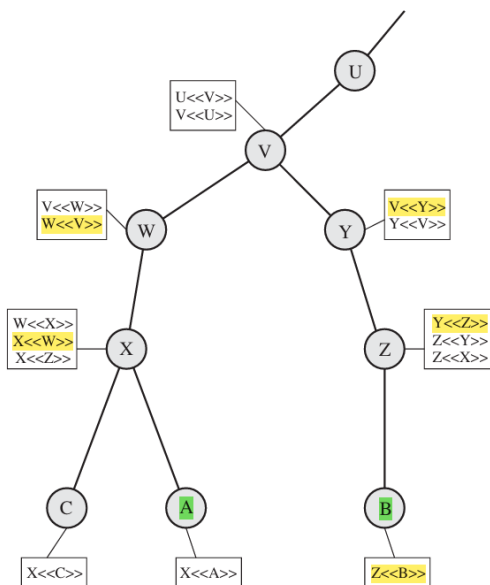
Ticket lifetime

Issuance of corresponding ticket against each different services

Qno. 06

Chain of trust

B will contact the CA2 which will further contact other CA's till CA1 is found and then a path construction will lead to final destination CA



(2)

Qno. 03

Given

$$p = 7$$

$$q = 11$$

$$e = 7$$

$$n = pq = 7 \times 11 = 77$$

$$\phi(n) = (p-1)(q-1) = 6 \times 10 = 60$$

$$d = ?$$

$$dx \bmod \phi(n) = dx \bmod 60 = 1$$

$$d = 43$$

$$43 \times 7 \bmod 60 = 1$$

$$301 \bmod 60 = 1$$

Now take cipher text $C1 = 10$

$$C = P^e \bmod n = 10^7 \bmod 77 = 10 \text{ (we got same number, RSA fails on small size input!!)}$$

Qno. 04

User A can validate both the integrity and authenticity of an operating system image downloaded from a vendor website using digital signatures and hashing.

Integrity – file is not modified on the way

Authenticity – file came from the genuine vendor

Vendors private key will be used to create hash of OS image, user will download the file with vendor's public key digitally signed by corresponding CA. Further, simple comparison will be made.

Qno. 05

By evaluating the Boolean condition:

$$\text{hash}(p3ach \mid 2287) == ab4jp$$

(3)

(A)

"I eat soup in Chicago Enigma"

I (8), E (4), S (18), I (8), C (2), E(4) = $8+4+18+8+2+4 = 44$

(B)

A=0, B=1, C=2,...

Say if we pick letters: B(1), C(2), D(3) = $1 \times 2 \times 3$

Changing one letter changes result drastically.

Reason:

Many combinations produce same product (factorization)

Different inputs can produce same result, for example if letter A is first, entire result will be 0, and frequency of A is quite higher in sentences.

(C)

Each word contributes now 2 letters, so more information is involved in digest.

Reduces the probability of collision, digest become a little bit stronger.

(D)

	Confidentiality	Integrity	Authentication	Non-repudiation
Sender +	Not used	Used to verify sign hash	Used to verify sender's identity	Used to verify sender cannot deny sending data
Sender -	Not used	Used to sign message digest	Used to prove sender's identity	Used to provide digital signature
Receiver +	Used by sender to encrypt data	Not used	Anyone can use it	Anyone can use it
Receiver -	Used to decrypt data	Not used	Used to receive message securely	Receiver can deny receiving